
MOSFET: Complete Study Guide

From Physics to Amplifiers — A Teacher-Style Reference

Topics Covered

- MOS Structure & Physics
- Channel Formation & V_{TH}
- Triode & Saturation Regions
- I-V Characteristics
- Transconductance g_m
- Diode-Connected MOSFETs
- Current Mirrors
- Common-Source Amplifier
- Small-Signal Models
- Design-Based Problems

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Razavi, “Fundamentals of Microelectronics”

Sedra/Smith, “Microelectronic Circuits”

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Contents

1	MOS Structure & Fundamentals	1
1.1	Why MOSFETs? — Motivation	1
1.2	Structure of the NMOS Transistor	1
1.2.1	The MOS Capacitor — Starting Point	1
1.2.2	Full MOSFET Structure	2
1.2.3	Key Dimensions	2
1.3	How Does the MOSFET Turn On? — Channel Formation	3
1.3.1	Step 1: Small V_G — Depletion Region	3
1.3.2	Step 2: $V_G \geq V_{TH}$ — Inversion Layer (Channel)	3
1.3.3	Step 3: MOSFET as a Variable Resistor	4
1.4	What Controls the Current? — Intuitive Understanding	4
1.5	Channel Pinch-Off — The Key to Saturation	5
1.5.1	How Pinch-Off Occurs	5
1.6	NMOS vs PMOS	6
1.7	MOSFET vs BJT — Quick Comparison	6
2	I-V Characteristics & Device Models	7
2.1	Derivation of the Drain Current	7
2.1.1	Step 1: Channel Charge Density	7
2.1.2	Step 2: From Charge to Current	8
2.2	Operating Regions	9
2.2.1	Cutoff Region	9
2.2.2	Triode (Linear) Region	9
2.2.3	Saturation Region	9
2.2.4	Quick Reference: How to Determine the Region	10
2.3	Channel-Length Modulation (λ)	10
2.4	MOS Transconductance g_m	11
2.4.1	Three Expressions for g_m	11
2.4.2	Understanding the Three Forms	11
2.5	Small-Signal Model	12
2.5.1	Basic Model ($\lambda = 0$)	12
2.5.2	Model with Channel-Length Modulation ($\lambda \neq 0$)	12
2.6	Large-Signal Models — Summary	13
2.7	Second-Order Effects (Brief Overview)	13
2.7.1	Body Effect	13
2.7.2	Velocity Saturation	13

3	MOSFET as a Diode (Diode-Connected Configuration)	15
3.1	What is a Diode-Connected MOSFET?	15
3.2	Why Does It Always Stay in Saturation?	16
3.3	I-V Characteristic of the Diode-Connected MOSFET	16
3.4	Small-Signal Resistance	16
3.5	Why NOT Use a Source-Connected Gate?	17
3.6	Applications of Diode-Connected MOSFETs	17
4	MOSFET Current Mirrors	19
4.1	The Need for Current Sources in ICs	19
4.2	Basic NMOS Current Mirror	19
4.2.1	Circuit Topology	19
4.2.2	How It Works — Teacher-Style Explanation	20
4.3	Setting the Reference Current	21
4.4	Effect of Channel-Length Modulation	22
4.5	Output Resistance of the Current Mirror	22
4.6	PMOS Current Mirror	23
4.7	Multiple Outputs and Current Scaling	23
4.8	Summary: Current Mirror Design Checklist	24
5	MOSFET Amplifiers (Common-Source)	25
5.1	The Common-Source (CS) Core	25
5.1.1	Basic CS Stage with Resistive Load	25
5.1.2	Small-Signal Analysis (Assuming $\lambda = 0$)	26
5.1.3	Input and Output Impedances	26
5.2	CS Stage with Channel-Length Modulation ($\lambda \neq 0$)	26
5.3	CS Stage with a Current-Source Load	27
5.4	CS Stage with Source Degeneration	28
5.4.1	Gain (Assuming $\lambda = 0$)	28
5.4.2	Output Resistance (with $\lambda \neq 0$)	29
5.4.3	Summary of CS Stages	29
6	Practice Problems & Design Challenges	31
6.1	Fundamentals & I-V Characteristics	31
6.2	Current Mirror Design	32
6.3	Amplifier Analysis	33

Chapter 1

MOS Structure & Fundamentals

1.1 Why MOSFETs? — Motivation

From BJT Limitations to MOSFET Innovation

Before diving into MOSFETs, let us understand **why** they were needed. The Bipolar Junction Transistor (BJT) has two key limitations:

1. **Emitter & Collector are connected through the Base** — charge carriers must pass through the base region, which limits control.
2. **Base Current $\Rightarrow$ Static Power Consumption** — the BJT always draws current at the input terminal, wasting power even when idle.

The MOSFET Solution:

1. Keep “Source” and “Drain” **physically separate** using oxide insulation — no direct path between them initially.
2. Use the **Field Effect** to create a conducting channel only when needed — the gate draws *zero* DC current because it sits on an insulator.

Key Takeaway: The gate of a MOSFET is electrically insulated from the rest of the device. This means **zero gate current** ($I_G = 0$), which is a massive advantage over the BJT in terms of power consumption and input impedance.

1.2 Structure of the NMOS Transistor

1.2.1 The MOS Capacitor — Starting Point

Think of it as a Capacitor First

Imagine the simplest possible structure:

- A **conductive plate** (metal or polysilicon) on top.
- A thin **insulator** (silicon dioxide, SiO_2) in the middle.

- A piece of **p-type silicon** (the substrate) at the bottom.

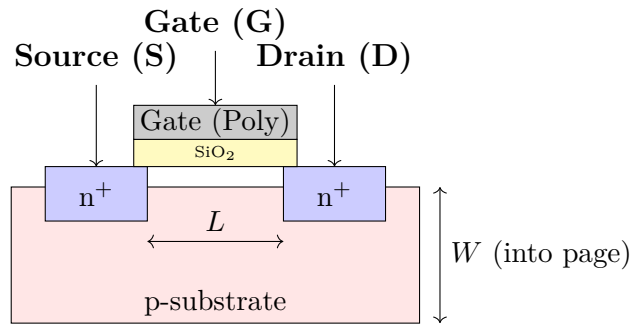
This is fundamentally a **capacitor**. When you put positive charge on the top plate, it attracts negative charge from the silicon below. The relationship is simply:

$$Q = C \cdot V$$

where C is the capacitance between the plates and V is the applied voltage. A higher C means better control of charge — this is why we want the oxide to be as **thin** as possible (today $\sim 15\text{--}18 \text{ \AA}$).

1.2.2 Full MOSFET Structure

The complete NMOS transistor adds two heavily-doped n-type regions (n^+) to the substrate:



Terminal Names:

- **Gate (G):** Controls the channel — acts as one plate of a capacitor.
- **Source (S):** Provides charge carriers (electrons in NMOS).
- **Drain (D):** Absorbs the charge carriers.
- The device is **symmetric** — Source and Drain are interchangeable depending on voltages.

1.2.3 Key Dimensions

Parameter	Symbol	Typical Value
Channel length	L	$\sim 90 \text{ nm}$ (designer-controlled)
Channel width	W	varies (designer-controlled)
Oxide thickness	t_{ox}	$\sim 15\text{--}18 \text{ \AA}$ (fixed by fabrication)

Table 1.1: Key MOSFET dimensions. Only W and L are under the circuit designer's control.

Designer's Parameters: W and L (lateral dimensions).

Technology Parameters: t_{ox} , μ_n , C_{ox} (vertical/material properties — fixed for all transistors on a chip).

1.3 How Does the MOSFET Turn On? — Channel Formation

This is the most important physical concept. Let us build it step by step.

1.3.1 Step 1: Small V_G — Depletion Region

What happens when you first apply a positive V_G ?

With Source, Drain, and Substrate all grounded, we increase V_G from zero:

1. The positive charge on the gate **repels holes** (majority carriers in p-type substrate) away from the surface.
2. This exposes **immobile negative ions** (the acceptor atoms that donated holes), creating a **depletion region** under the gate.
3. The device acts like a capacitor: positive charge on the gate is mirrored by negative charge (ions) in the substrate.
4. **But there is no channel yet!** No mobile carriers exist at the surface, so **no current** can flow from Source to Drain. The MOSFET is **OFF**.

1.3.2 Step 2: $V_G \geq V_{TH}$ — Inversion Layer (Channel)

The Threshold Voltage and Channel Formation

As V_G continues to increase:

1. The electric field pointing into the substrate gets stronger and stronger.
2. At some critical voltage called the **threshold voltage** V_{TH} (typically 300–500 mV), the field is strong enough to attract **free electrons** from the n^+ Source and Drain regions to the oxide-silicon interface.
3. A thin layer of electrons forms at the surface — this is the **inversion layer** (the p-type surface has been “inverted” to n-type behavior).
4. This electron layer creates a **conducting channel** between Source and Drain.
5. The MOSFET is now **ON**.

The channel charge density increases as V_G rises above V_{TH} :

$$Q_{\text{channel}} \propto C_{ox} \cdot (V_G - V_{TH})$$

More voltage $\Rightarrow$ more electrons $\Rightarrow$ lower resistance $\Rightarrow$ more current capability.

Threshold Voltage V_{TH} : The minimum gate-source voltage needed to form a conducting channel.

$$\boxed{\text{MOSFET is ON when } V_{GS} \geq V_{TH}}$$

1.3.3 Step 3: MOSFET as a Variable Resistor

Once the channel exists ($V_{GS} > V_{TH}$), the Source-to-Drain path behaves like a **resistor** whose value is controlled by V_G :

- Higher $V_G \Rightarrow$ denser channel $\Rightarrow$ lower resistance.
- $V_G = V_{TH} \Rightarrow$ channel just barely exists $\Rightarrow R_{on} = \infty$ (device is OFF).

This voltage-dependent resistance is the foundation of MOSFET operation in both analog and digital circuits.

On-Resistance (deep triode / small V_{DS}):

$$\boxed{R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})}}$$

1.4 What Controls the Current? — Intuitive Understanding

Factors Affecting Drain Current

The drain current I_D depends on several factors. Think about each one intuitively:

1. **Gate Voltage (V_{GS}):** More voltage above $V_{TH} \Rightarrow$ more electrons in channel $\Rightarrow$ more current.
2. **Channel Length (L):** Longer channel $\Rightarrow$ more resistance $\Rightarrow$ less current.
(Like a longer garden hose — harder to push water through.)
3. **Channel Width (W):** Wider channel $\Rightarrow$ more cross-section for flow $\Rightarrow$ more current.
(Like opening more lanes on a highway.)
A device with width $2W$ is equivalent to two devices of width W in parallel.
4. **Oxide Thickness (t_{ox}):** Thinner oxide $\Rightarrow$ larger capacitance ($C_{ox} = \epsilon_{ox}/t_{ox}$) $\Rightarrow$ more charge for same $V_G \Rightarrow$ more current.
5. **Electron Mobility (μ_n):** Higher mobility $\Rightarrow$ electrons move faster $\Rightarrow$ more current. Mobility decreases at high temperatures.
6. **Drain Voltage (V_{DS}):** Creates the electric field that drives current through the channel.

Transport Mechanism: Current in a MOSFET is due to **drift** (electric-field-driven), not diffusion. The drain voltage creates a horizontal electric field along the channel that sweeps electrons from Source to Drain.

1.5 Channel Pinch-Off — The Key to Saturation

This is the **most important concept** for understanding why the MOSFET can act as a current source.

1.5.1 How Pinch-Off Occurs

Step-by-Step Explanation of Pinch-Off

Setup: $V_{GS} > V_{TH}$ (channel exists), and we gradually increase V_{DS} .

1. The channel voltage is **not uniform**: it varies from 0 V at the Source end to V_{DS} at the Drain end (because current flows through the channel resistance).
2. To sustain a channel at any point x , the local potential difference between Gate and that point must exceed V_{TH} :

$$V_G - V(x) \geq V_{TH}$$

3. At the **Source end**: $V(0) = 0$, so the condition is $V_{GS} \geq V_{TH}$ ✓ (always met).
4. At the **Drain end**: $V(L) = V_{DS}$, so the condition becomes:

$$V_{GS} - V_{DS} \geq V_{TH} \quad \Rightarrow \quad V_{DS} \leq V_{GS} - V_{TH}$$

5. When $V_{DS} = V_{GS} - V_{TH}$: the channel **just barely disappears** at the Drain end. This is called **pinch-off**.
6. When $V_{DS} > V_{GS} - V_{TH}$: the channel ends before reaching the Drain. But **current does NOT stop!** Electrons reaching the end of the channel are swept to the Drain by the strong electric field in the depletion region.

[Image: MOSFET Channel Formation]

Figure 1.1: Illustration of channel pinch-off as V_{DS} increases.

Critical Insight: After pinch-off, further increases in V_{DS} do **not** significantly increase the current — the MOSFET acts as a **constant current source**. This is the **saturation region**.

Pinch-off condition: $V_{DS} = V_{GS} - V_{TH}$ (also written as $V_{OV} = V_{GS} - V_{TH}$)

1.6 NMOS vs PMOS

Property	NMOS	PMOS
Substrate	p-type	n-type (n-well)
Source/Drain	n^+	p^+
Channel carriers	Electrons	Holes
Turn-on condition	$V_{GS} > V_{TH}$ ($V_{TH} > 0$)	$V_{GS} < V_{TH}$ ($V_{TH} < 0$)
Saturation condition	$V_{DS} > V_{GS} - V_{TH}$	$V_{DS} < V_{GS} - V_{TH}$
Mobility	μ_n (higher)	μ_p (lower)
Performance	Better (preferred)	Worse (slower for same size)

Table 1.2: NMOS vs PMOS comparison.

CMOS Technology: Both NMOS and PMOS can be fabricated on the same chip by creating an “n-well” in the p-substrate for the PMOS devices. This complementary MOS (CMOS) technology is the foundation of all modern digital and analog ICs.

1.7 MOSFET vs BJT — Quick Comparison

Property	BJT	MOSFET
I-V relationship	Exponential: $I_C \propto e^{V_{BE}/V_T}$	Square-law: $I_D \propto (V_{GS} - V_{TH})^2$
Edge of saturation	$V_{CE} = V_{BE}$ (i.e., $V_{CB} = 0$)	$V_{DS} = V_{GS} - V_{TH}$
Input current	Finite $I_B = I_C/\beta$	Zero: $I_G = 0$
g_m for same I	Higher (I_C/V_T)	Lower ($2I_D/V_{OV}$)
Device sizing	Same dimensions, same I_S	Each device sized individually
Output effect	Early effect (V_A)	Channel-length modulation (λ)
Current mechanism	Diffusion	Drift

Table 1.3: BJT vs MOSFET comparison.

Chapter 2

I-V Characteristics & Device Models

2.1 Derivation of the Drain Current

The Big Picture

We want to find how the drain current I_D depends on:

- Gate-source voltage V_{GS}
- Drain-source voltage V_{DS}
- Device parameters (μ_n , C_{ox} , W/L)

The derivation uses two simple ideas: (1) find the charge in the channel, and (2) figure out how fast it moves.

2.1.1 Step 1: Channel Charge Density

Finding the Charge per Unit Length $Q(x)$

The gate-channel system is a capacitor. The charge per unit length is:

$$Q = C \times V$$

where:

- $C = W \cdot C_{ox}$ is the gate capacitance **per unit length** (in F/m). We multiply C_{ox} (F/m²) by W to get capacitance per unit length.
- $V = V_{GS} - V_{TH}$ is the effective voltage creating the channel (no charge exists below V_{TH}).

Near the **Source end** (where channel voltage ≈ 0):

$$Q = WC_{ox}(V_{GS} - V_{TH})$$

But the channel voltage varies along the length! At position x , the channel sits at voltage $V(x)$, so the **local** charge density is:

$$Q(x) = WC_{ox}[V_{GS} - V(x) - V_{TH}]$$

Note: $V(x)$ goes from 0 at Source to V_{DS} at Drain.

2.1.2 Step 2: From Charge to Current

Relating Current to Charge and Velocity

Current = charge per unit length $\times$ velocity of carriers:

$$I_D = Q(x) \cdot v(x)$$

The velocity is related to the electric field by the **drift** equation:

$$v = \mu_n E = \mu_n \frac{dV}{dx}$$

Substituting:

$$I_D = WC_{ox}[V_{GS} - V(x) - V_{TH}] \cdot \mu_n \frac{dV(x)}{dx}$$

Key insight: I_D must be the **same** at every point along the channel (conservation of charge — current in = current out). So we can integrate both sides:

$$\begin{aligned} \int_0^L I_D dx &= \int_0^{V_{DS}} \mu_n C_{ox} W [V_{GS} - V(x) - V_{TH}] dV \\ I_D \cdot L &= \mu_n C_{ox} W \left[(V_{GS} - V_{TH})V_{DS} - \frac{V_{DS}^2}{2} \right] \end{aligned}$$

General I-V Equation (Triode Region):

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left[2(V_{GS} - V_{TH})V_{DS} - V_{DS}^2 \right]$$

Valid for $V_{GS} \geq V_{TH}$ and $V_{DS} \leq V_{GS} - V_{TH}$.

2.2 Operating Regions

2.2.1 Cutoff Region

Cutoff: $V_{GS} < V_{TH} \implies I_D = 0$
No channel exists. The transistor is OFF.

2.2.2 Triode (Linear) Region

Condition: $V_{GS} \geq V_{TH}$ and $V_{DS} < V_{GS} - V_{TH}$ (equivalently $V_{GD} > V_{TH}$).

Triode Region:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

Deep Triode ($V_{DS} \ll 2(V_{GS} - V_{TH})$): the $V_{DS}^2/2$ term is negligible:

Deep Triode (Resistive):

$$I_D \approx \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) V_{DS} \quad \Rightarrow \quad R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})}$$

The MOSFET acts as a **linear voltage-controlled resistor**.

2.2.3 Saturation Region

Condition: $V_{GS} \geq V_{TH}$ and $V_{DS} \geq V_{GS} - V_{TH}$ (equivalently $V_{GD} \leq V_{TH}$).

At pinch-off, $V_{DS} = V_{GS} - V_{TH}$. Substituting into the triode equation:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH})^2 - \frac{(V_{GS} - V_{TH})^2}{2} \right] = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

Saturation Region (without λ):

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

The current is **independent of V_{DS}** — the MOSFET is a **current source** controlled by V_{GS} .

The quantity $V_{OV} = V_{GS} - V_{TH}$ is called the **overdrive voltage**.

2.2.4 Quick Reference: How to Determine the Region

Simple Decision Tree

1. Is $V_{GS} \geq V_{TH}$?

No $\Rightarrow$ **Cutoff** ($I_D = 0$).

Yes $\Rightarrow$ Go to step 2.

2. Is $V_{DS} \geq V_{GS} - V_{TH}$ (i.e., $V_{GD} \leq V_{TH}$)?

Yes $\Rightarrow$ **Saturation**.

No $\Rightarrow$ **Triode**.

Tip: If the drain voltage is close to (but higher than) the gate voltage, the MOSFET is almost certainly saturated.

2.3 Channel-Length Modulation (λ)

Why the Current Isn't Perfectly Flat in Saturation

In reality, as V_{DS} increases beyond pinch-off:

- The pinch-off point moves slightly toward the Source.
- The effective channel length L_{eff} **decreases**.
- Since $I_D \propto 1/L$, the current **increases slightly**.

This is called **channel-length modulation** — the MOS equivalent of the Early effect in BJTs.

We model this by adding a correction factor $(1 + \lambda V_{DS})$:

Saturation with Channel-Length Modulation:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$$

where λ is the channel-length modulation coefficient (units: V^{-1}).

Key facts about λ :

- $\lambda \propto 1/L$ — longer channels have less modulation (designer-controlled!).
- Unlike the BJT Early voltage, λ can be adjusted by choosing L .

2.4 MOS Transconductance g_m

What is g_m ?

Transconductance measures how effectively the gate voltage controls the drain current:

$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

A higher g_m means the device is a “stronger” amplifier — a small change in V_{GS} produces a large change in I_D .

2.4.1 Three Expressions for g_m

Starting from $I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$, we can derive three equivalent forms:

Derivation of the Three g_m Expressions

Form 1: Differentiate directly:

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) \quad (1)$$

Form 2: From $I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$, solve for $(V_{GS} - V_{TH}) = \sqrt{2I_D / (\mu_n C_{ox} W/L)}$ and substitute into (1):

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} \cdot I_D} \quad (2)$$

Form 3: Divide (1) by the saturation current equation:

$$\frac{g_m}{I_D} = \frac{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})}{\frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2} = \frac{2}{V_{GS} - V_{TH}}$$

$$g_m = \frac{2I_D}{V_{GS} - V_{TH}} \quad (3)$$

2.4.2 Understanding the Three Forms

Form	Equation	What's Fixed	g_m scales with
(1)	$\mu_n C_{ox} \frac{W}{L} \cdot V_{OV}$	W/L fixed, vary V_{OV}	$g_m \propto V_{OV}$
(2)	$\sqrt{2\mu_n C_{ox} \frac{W}{L} \cdot I_D}$	W/L fixed, vary I_D	$g_m \propto \sqrt{I_D}$
(3)	$2I_D/V_{OV}$	I_D fixed, vary W/L	$g_m \propto 1/V_{OV}$

Table 2.1: Dependencies of g_m — seemingly contradictory but each applies under different constraints.

Why do Forms (1) and (2) seem contradictory?

Form (1) says $g_m \propto V_{OV}$ (increases with overdrive).

Form (3) says $g_m \propto 1/V_{OV}$ (decreases with overdrive).

The resolution: in (1), W/L is fixed, so increasing V_{OV} also increases I_D quadratically. In (3), I_D is fixed, so increasing V_{OV} requires **decreasing** W/L , which reduces g_m .

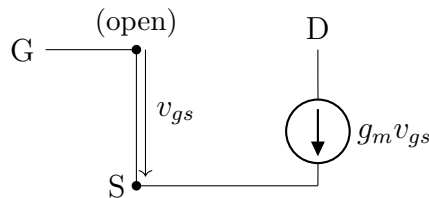
They are both correct — just under different constraints!

2.5 Small-Signal Model

From Large-Signal to Small-Signal

When the MOSFET is biased in saturation with a DC operating point, and we apply a **small** AC signal on top, the nonlinear I-V relationship can be **linearized** around the operating point. This gives us a simple linear model for AC analysis.

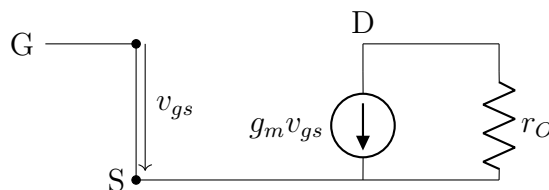
2.5.1 Basic Model ($\lambda = 0$)



- The gate is an **open circuit** (zero gate current).
- The drain current is a **voltage-controlled current source**: $i_d = g_m v_{gs}$.

2.5.2 Model with Channel-Length Modulation ($\lambda \neq 0$)

Add a resistor r_O between Drain and Source to model the finite output impedance:



Output Resistance:

$$r_O = \left(\frac{\partial I_D}{\partial V_{DS}} \right)^{-1} \approx \frac{1}{\lambda I_D}$$

Intrinsic Gain (maximum voltage gain from a single transistor):

$$A_v = g_m r_O = \frac{2}{\lambda (V_{GS} - V_{TH})} = \frac{2}{\lambda V_{OV}}$$

Example: Small-Signal Parameters

A MOSFET has $I_D = 0.5 \text{ mA}$, $\mu_n C_{ox} = 100 \text{ } \mu\text{A/V}^2$, $W/L = 10$, $\lambda = 0.1 \text{ V}^{-1}$.

Find g_m and r_O :

$$g_m = \sqrt{2 \times 100 \times 10^{-6} \times 10 \times 0.5 \times 10^{-3}} = \sqrt{10^{-3}} = \frac{1}{1 \text{ k}\Omega} \approx 1 \text{ mS}$$

$$r_O = \frac{1}{\lambda I_D} = \frac{1}{0.1 \times 0.5 \times 10^{-3}} = 20 \text{ k}\Omega$$

Intrinsic gain: $g_m r_O = 1 \text{ mS} \times 20 \text{ k}\Omega = 20$

2.6 Large-Signal Models — Summary**Complete NMOS Model Summary:**

Cutoff ($V_{GS} < V_{TH}$): $I_D = 0$

Triode ($V_{GS} \geq V_{TH}$, $V_{DS} < V_{OV}$): $I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH})V_{DS} - \frac{V_{DS}^2}{2} \right]$

Saturation ($V_{GS} \geq V_{TH}$, $V_{DS} \geq V_{OV}$): $I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$

Deep Triode ($V_{DS} \ll 2V_{OV}$): $I_D \approx \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})V_{DS}$

Complete PMOS Model Summary (using absolute values):

Cutoff ($|V_{GS}| < |V_{TH}|$): $|I_D| = 0$

Saturation ($|V_{DS}| \geq |V_{GS}| - |V_{TH}|$): $|I_D| = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (|V_{GS}| - |V_{TH}|)^2 (1 + \lambda |V_{DS}|)$

Triode ($|V_{DS}| < |V_{GS}| - |V_{TH}|$): $|I_D| = \mu_p C_{ox} \frac{W}{L} \left[(|V_{GS}| - |V_{TH}|)|V_{DS}| - \frac{V_{DS}^2}{2} \right]$

2.7 Second-Order Effects (Brief Overview)**2.7.1 Body Effect**

When the Source is not at the same potential as the Substrate (Body), the threshold voltage changes:

$$V_{TH} = V_{TH0} + \gamma \left(\sqrt{|2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|} \right)$$

where $\gamma \approx 0.4 \sqrt{\text{V}}$ and $\phi_F \approx 0.4 \text{ V}$ are technology parameters. As V_{SB} increases, V_{TH} increases, reducing the current.

2.7.2 Velocity Saturation

In very short channels ($L < 0.1 \text{ } \mu\text{m}$), the electric field is so strong that carrier velocity saturates at v_{sat} . The current becomes:

$$I_D = v_{sat} \cdot W C_{ox} (V_{GS} - V_{TH})$$

This is **linear** in V_{OV} (not quadratic), and $g_m = v_{sat} W C_{ox}$ becomes independent of L and I_D .

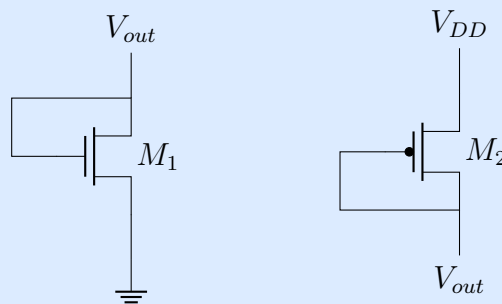
Chapter 3

MOSFET as a Diode (Diode-Connected Configuration)

3.1 What is a Diode-Connected MOSFET?

The Simple Idea

A **diode-connected MOSFET** is created by connecting the **gate to the drain**:



Left: NMOS diode-connected. Right: PMOS diode-connected.

[Image: Diode-Connected MOSFET]

Figure 3.1: Razavi's visual reference for the diode-connected configuration.

Why “diode-connected”? Just like a diode is a two-terminal device that conducts in one direction, this configuration turns the three-terminal MOSFET into a **two-terminal device**.

Immediately obvious: Since $V_G = V_D$, we have $V_{GD} = 0$. Since $V_{TH} > 0$ for NMOS, this means $V_{GD} < V_{TH}$, so the device is **always in saturation** (as long as it is on). This is a very important property!

3.2 Why Does It Always Stay in Saturation?

Proof: Diode-Connected MOSFET is Always Saturated

For saturation, we need: $V_{DS} \geq V_{GS} - V_{TH}$.

With the gate tied to the drain: $V_{GS} = V_{DS}$.

Substituting: $V_{GS} \geq V_{GS} - V_{TH}$, which gives $V_{TH} \geq 0$.

Since $V_{TH} > 0$ for an NMOS device, this is **always true**. Therefore the device is **always in saturation** whenever it is on ($V_{GS} > V_{TH}$). $\square$

3.3 I-V Characteristic of the Diode-Connected MOSFET

Since the device is always saturated, and $V_{DS} = V_{GS}$:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

This is a **square-law** relationship between I_D and V (the voltage across the two terminals). Compare this to a regular diode which has an *exponential* I-V relationship.

Diode-Connected MOSFET I-V:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V - V_{TH})^2$$

where V is the voltage across the device (drain-to-source = gate-to-source).
Conversely, the voltage for a given current is:

$$V = V_{TH} + \sqrt{\frac{2I_D}{\mu_n C_{ox} \frac{W}{L}}}$$

3.4 Small-Signal Resistance

What Does It Look Like to Small Signals?

This is the most useful property of the diode-connected MOSFET. To a small AC signal, it looks like a **resistor**. Let us find its value.

Deriving the Small-Signal Resistance

Apply a test voltage v_x and measure the current i_x .
In the small-signal model:

- Since gate is tied to drain, $v_{gs} = v_x$.
- The current source produces $g_m v_{gs} = g_m v_x$.
- If $\lambda \neq 0$, the output resistance r_O is in parallel.

The total current is:

$$i_x = g_m v_x + \frac{v_x}{r_O}$$

Therefore:

$$R_{\text{diode}} = \frac{v_x}{i_x} = \frac{1}{g_m + 1/r_O} = \frac{1}{g_m} \parallel r_O$$

Since $r_O \gg 1/g_m$ in practice:

$$R_{\text{diode}} \approx \frac{1}{g_m}$$

Small-Signal Resistance of Diode-Connected MOSFET:

$$R_{\text{diode}} = \frac{1}{g_m} \parallel r_O \approx \frac{1}{g_m}$$

Using the three forms of g_m :

$$\frac{1}{g_m} = \frac{1}{\mu_n C_{ox} \frac{W}{L} V_{OV}} = \frac{1}{\sqrt{2\mu_n C_{ox} \frac{W}{L} \cdot I_D}} = \frac{V_{OV}}{2I_D}$$

Contrast with BJT: A diode-connected BJT has resistance $1/g_m = V_T/I_C \approx 26 \text{ mV}/I_C$. A diode-connected MOSFET has resistance $V_{OV}/(2I_D)$. Since V_{OV} (typically 100–300 mV) $\gg V_T$ (26 mV), the MOSFET diode has a **higher resistance** for the same current.

3.5 Why NOT Use a Source-Connected Gate?

Common Mistake to Avoid

Students sometimes ask: “Why not connect the gate to the *source* instead?”

If you connect the gate to the source, V_{GS} becomes **fixed at zero**, and the MOSFET is permanently OFF (since $0 < V_{TH}$). This configuration is useless.

The gate must be connected to the **drain** so that $V_{GS} = V_{DS}$ varies with the signal, keeping the device on and in saturation.

3.6 Applications of Diode-Connected MOSFETs

1. **Load in amplifiers:** Used as a drain load in CS stages (provides resistance $1/g_m$).
2. **Reference element in current mirrors:** The “master” side of a current mirror is a diode-connected device.

3. **Voltage shifting / level setting:** The voltage drop $V_{GS} = V_{TH} + V_{OV}$ provides a defined voltage shift.
4. **Active loads:** Replace passive resistors in integrated circuits where large resistances are impractical.

Example: CS Stage with Diode-Connected Load

Consider a common-source stage where the load R_D is replaced by a diode-connected NMOS M_2 :

The gain becomes:

$$A_v = -g_m[1] \times R_{\text{load}} = -g_m[1] \times \frac{1}{g_m[2]} = -\frac{g_m[1]}{g_m[2]}$$

Using $g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} \cdot I_D}$ and noting both carry the same I_D :

$$A_v = -\sqrt{\frac{(W/L)_1}{(W/L)_2}}$$

Key insight: The gain depends **only on geometry ratios**, not on $\mu_n C_{ox}$ or I_D . This makes the circuit very robust against process and temperature variations!

Contrast: A bipolar version would give $A_v = -g_{m1}/g_{m2} = -I_{C1}/I_{C2} \approx -1$ (useless gain).

Chapter 4

MOSFET Current Mirrors

4.1 The Need for Current Sources in ICs

Why Current Sources?

In integrated circuits, we need **constant bias currents** for amplifiers and other analog blocks. Resistors are impractical on-chip because:

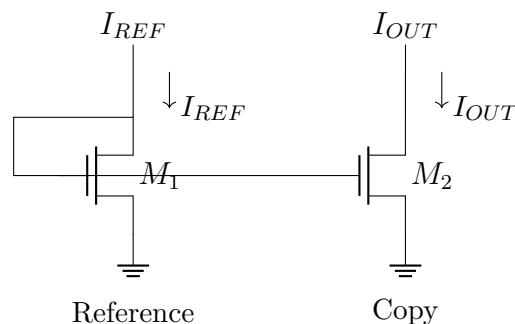
- Large resistors consume too much chip area.
- Resistor values are inaccurate ($\pm 20\%$ tolerance).

Instead, we use MOSFETs biased in **saturation** as current sources. In saturation, I_D is (ideally) independent of V_{DS} — exactly what a current source should be.

But how do we set the current to a precise value? Answer: **Current Mirrors**.

4.2 Basic NMOS Current Mirror

4.2.1 Circuit Topology



[Image: Current Mirror p.80]

Figure 4.1: Sedra's basic current mirror topology.

4.2.2 How It Works — Teacher-Style Explanation

Step-by-Step Understanding

1. **The reference side (M_1):** M_1 is diode-connected (gate tied to drain). A known reference current I_{REF} is forced through it (by an external circuit or resistor to V_{DD}).

2. M_1 **sets** V_{GS} : Since M_1 is in saturation (always, because it's diode-connected):

$$I_{REF} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_{GS} - V_{TH})^2$$

This equation **uniquely determines** V_{GS} for the given I_{REF} .

3. **Gates are shared:** M_1 and M_2 have the **same gate voltage** (they're connected). If both have the same V_{TH} (same technology), they have the **same** V_{GS} .

4. M_2 **copies the current:** If M_2 is also in saturation:

$$I_{OUT} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_2 (V_{GS} - V_{TH})^2$$

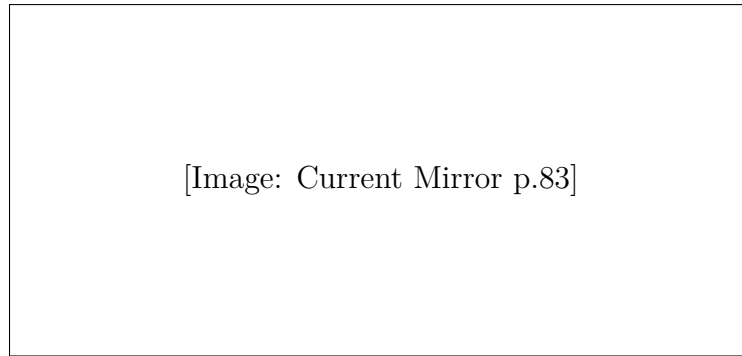
5. **The ratio:** Dividing:

$$\frac{I_{OUT}}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$

Basic Current Mirror Equation:

$$I_{OUT} = I_{REF} \cdot \frac{(W/L)_2}{(W/L)_1}$$

- If M_1 and M_2 are **identical** (W/L the same): $I_{OUT} = I_{REF}$ (exact copy).
- If $(W/L)_2 = 2(W/L)_1$: $I_{OUT} = 2 \cdot I_{REF}$ (scaled copy).
- The current can be **scaled up or down** by choosing device ratios.



[Image: Current Mirror p.83]

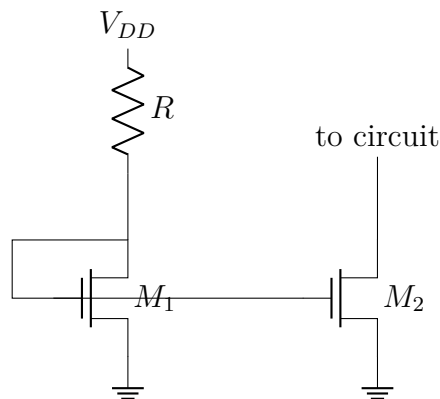
Figure 4.2: Sedra’s mathematical derivation of the current mirror ratio and output resistance effect.

Key Assumptions:

- Both transistors have the **same** V_{TH} (guaranteed if they’re on the same chip).
- Both are in **saturation** (M_2 must have sufficient V_{DS}).
- $\lambda = 0$ (no channel-length modulation) — we’ll relax this soon.

4.3 Setting the Reference Current

The reference current I_{REF} is typically set using a resistor from V_{DD} :



Finding I_{REF}

KVL from V_{DD} through R and M_1 to ground:

$$V_{DD} = I_{REF} \cdot R + V_{GS}$$

And since M_1 is saturated:

$$I_{REF} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_{GS} - V_{TH})^2$$

These two equations can be solved simultaneously (by substitution or iteration) to find I_{REF} and V_{GS} .

Quick estimate: If V_{GS} is known approximately (e.g., $\sim 0.7\text{--}1\text{ V}$), then:

$$I_{REF} \approx \frac{V_{DD} - V_{GS}}{R}$$

4.4 Effect of Channel-Length Modulation

The Real-World Problem

In practice, M_1 and M_2 operate at **different** V_{DS} values:

- M_1 : $V_{DS1} = V_{GS}$ (because it's diode-connected).
- M_2 : V_{DS2} depends on the circuit connected to its drain.

If $\lambda \neq 0$, the currents become:

$$I_{REF} = \frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)_1 (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS1})$$

$$I_{OUT} = \frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)_2 (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS2})$$

Current Mirror with λ :

$$\frac{I_{OUT}}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1} \cdot \frac{1 + \lambda V_{DS2}}{1 + \lambda V_{DS1}}$$

The mismatch factor $\frac{1 + \lambda V_{DS2}}{1 + \lambda V_{DS1}}$ causes the output current to deviate from the ideal ratio.

How to minimize the mismatch:

1. Use **longer channels** (larger L) $\Rightarrow$ smaller λ .
2. Try to keep $V_{DS1} \approx V_{DS2}$ (cascode mirrors help).
3. The output resistance of the mirror (M_2) is $r_O = 1/(\lambda I_D)$ — finite, not infinite.

4.5 Output Resistance of the Current Mirror

How Good Is the Current Source?

An ideal current source has **infinite output resistance** (current doesn't change regardless of voltage). A real MOS current source has finite output resistance due to channel-length modulation.

For the basic mirror, M_2 has its gate at a fixed voltage (set by M_1). Therefore, M_2 is just a common-source transistor with a fixed V_{GS} :

Output Resistance of Basic Current Mirror:

$$R_{out} = r_O[2] = \frac{1}{\lambda I_{OUT}}$$

Numerical Example: Current Mirror Design

Given: $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$, $V_{TH} = 0.4 \text{ V}$, $\lambda = 0.1 \text{ V}^{-1}$, $V_{DD} = 1.8 \text{ V}$.

Design a current mirror to produce $I_{OUT} = 200 \mu\text{A}$ with identical transistors.

Step 1: Choose $I_{REF} = I_{OUT} = 200 \mu\text{A}$ (identical devices).

Step 2: Choose $V_{OV} = V_{GS} - V_{TH}$. Let's pick $V_{OV} = 0.3 \text{ V}$, so $V_{GS} = 0.7 \text{ V}$.

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \cdot V_{OV}^2 \Rightarrow 200 = \frac{1}{2} (200) \frac{W}{L} (0.3)^2$$

$$\frac{W}{L} = \frac{200 \times 2}{200 \times 0.09} = 22.2 \Rightarrow W/L = 22.2$$

Step 3: Reference resistor: $R = \frac{V_{DD} - V_{GS}}{I_{REF}} = \frac{1.8 - 0.7}{0.2 \text{ mA}} = 5.5 \text{ k}\Omega$

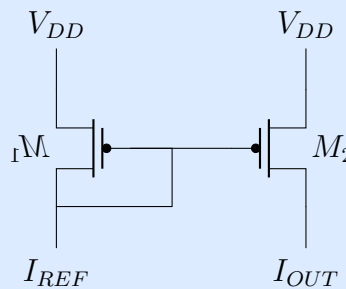
Step 4: Output resistance: $R_{out} = \frac{1}{\lambda I_{OUT}} = \frac{1}{0.1 \times 0.2 \times 10^{-3}} = 50 \text{ k}\Omega$

Step 5: Minimum V_{DS} for saturation: $V_{DS,min} = V_{OV} = 0.3 \text{ V}$.

4.6 PMOS Current Mirror

PMOS Version

The PMOS current mirror is the “mirror image” of the NMOS version. It connects to V_{DD} and **sources** current (pushes current down into a circuit):

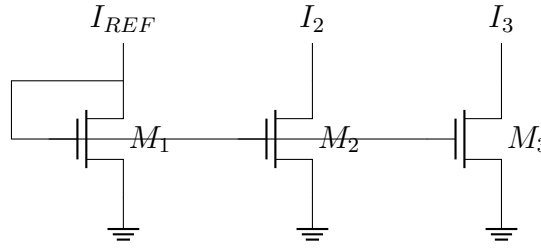


The same ratio applies: $I_{OUT}/I_{REF} = (W/L)_2/(W/L)_1$.

PMOS mirrors are extremely common as **active loads** for NMOS amplifier stages.

4.7 Multiple Outputs and Current Scaling

A single reference device can drive **multiple** mirror transistors:



$$I_2 = I_{REF} \cdot \frac{(W/L)_2}{(W/L)_1}, \quad I_3 = I_{REF} \cdot \frac{(W/L)_3}{(W/L)_1}$$

Design Flexibility: You can generate multiple different currents from a single reference. For example, if $(W/L)_1 = 10$, $(W/L)_2 = 20$, $(W/L)_3 = 5$, and $I_{REF} = 100 \mu\text{A}$, then $I_2 = 200 \mu\text{A}$ and $I_3 = 50 \mu\text{A}$.

4.8 Summary: Current Mirror Design Checklist

Design Steps:

1. Choose the desired I_{OUT} and V_{OV} (trade-off: smaller V_{OV} = less headroom consumed, but larger W/L needed).
2. Calculate W/L from $I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} V_{OV}^2$.
3. Set I_{REF} through a resistor or another current source.
4. Scale $(W/L)_2/(W/L)_1$ for the desired current ratio.
5. Verify saturation: ensure $V_{DS} \geq V_{OV}$ for all mirror transistors.
6. Check output resistance: $R_{out} = 1/(\lambda I_{OUT})$. If too low, use longer L or a cascode.

Chapter 5

MOSFET Amplifiers (Common-Source)

5.1 The Common-Source (CS) Core

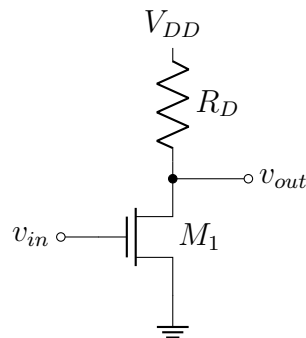
What is a Common-Source Amplifier?

A common-source (CS) stage is the most fundamental MOS amplifier.

- **Input** is applied to the **Gate**.
- **Output** is taken from the **Drain**.
- The **Source** is (AC) grounded — hence “Common-Source”.

It is the MOS equivalent of the Common-Emitter (CE) amplifier. The transistor acts as a voltage-controlled current source: it turns the small input voltage variations (v_{in}) into current variations ($i_d = g_m v_{in}$), which then flow through a load to create a large output voltage (v_{out}).

5.1.1 Basic CS Stage with Resistive Load



5.1.2 Small-Signal Analysis (Assuming $\lambda = 0$)

Deriving the Gain

In the small-signal model, V_{DD} is an AC ground.

- $v_{gs} = v_{in}$
- The drain current is $i_d = g_m v_{gs} = g_m v_{in}$
- This current flows *up* through R_D to AC ground, so the voltage at the drain is:

$$v_{out} = -i_d R_D = -g_m v_{in} R_D$$

Therefore, the voltage gain is:

$$A_v = \frac{v_{out}}{v_{in}} = -g_m R_D$$

The Minus Sign: The negative sign indicates a **180-degree phase shift** (inversion). When the input voltage goes up, the current goes up, which increases the voltage drop across R_D , pulling the output voltage *down*.

5.1.3 Input and Output Impedances

Input Impedance: Since the gate draws no low-frequency current, $R_{in} = \infty$. This is a major advantage over the BJT CE stage (where $R_{in} = r_\pi$).

Output Impedance: Looking into the drain, we see R_D in parallel with the transistor's output resistance. If $\lambda = 0$, the transistor's r_O is infinite, so $R_{out} = R_D$.

5.2 CS Stage with Channel-Length Modulation ($\lambda \neq 0$)

Including r_O

If $\lambda \neq 0$, the MOSFET has a finite output resistance $r_O = 1/(\lambda I_D)$. In the small-signal model, r_O appears between the drain and source. Since the source is grounded, r_O is in parallel with R_D .

Gain and Impedances with $\lambda \neq 0$:

$$A_v = -g_m (R_D \parallel r_O)$$

$$R_{in} = \infty, \quad R_{out} = R_D \parallel r_O$$

Example: CS Gain Limit

What is the maximum possible gain of a CS stage?

If we replace the resistor R_D with an ideal current source (which has infinite AC resistance), then $R_D \rightarrow \infty$.

The gain becomes:

$$A_v = -g_m r_O$$

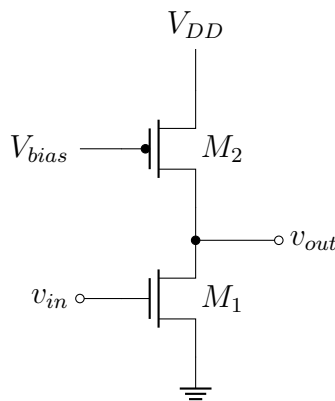
This is the **intrinsic gain** of the transistor. It is the absolute maximum voltage gain a single MOSFET can provide.

5.3 CS Stage with a Current-Source Load

Replacing the Resistor

To get high gain, we need a large load resistance. But a large R_D requires a huge V_{DD} to sustain the DC current ($V_{drop} = I_D R_D$).

Solution: Replace R_D with a **PMOS Current Source**. A current source has a small DC voltage drop but a very large AC resistance (r_O).

**Small-Signal Analysis**

- M_1 (NMOS) is the amplifying device. Its transconductance is $g_m[1]$ and its output resistance is $r_O[1]$.
- M_2 (PMOS) acts as the load. Its gate is at a constant DC voltage V_{bias} , so its AC $v_{gs2} = 0$. Therefore, its current source $g_m[2]v_{gs2} = 0$.
- M_2 simply looks like a resistor of value $r_O[2]$ to AC signals.
- The total load resistance seen by M_1 is $r_O[1] \parallel r_O[2]$.

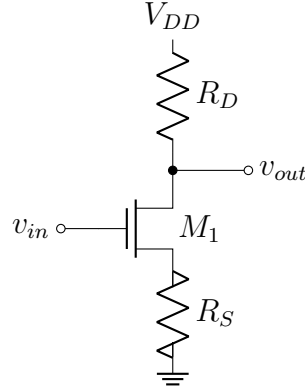
$$A_v = -g_m[1](r_O[1] \parallel r_O[2])$$

$$R_{out} = r_O[1] \parallel r_O[2]$$

5.4 CS Stage with Source Degeneration

Adding a Resistor at the Source

Placing a resistor R_S between the source and ground provides **negative feedback**. When v_{in} increases, i_d increases, which increases the voltage drop across R_S . This raises the source voltage, effectively reducing v_{gs} and fighting the increase in current.



5.4.1 Gain (Assuming $\lambda = 0$)

Deriving the Degenerated Gain

1. The input voltage is split between the gate-source junction and R_S :

$$v_{in} = v_{gs} + i_d R_S$$

2. Since $i_d = g_m v_{gs}$, we have $v_{gs} = i_d / g_m$. Substitute this:

$$v_{in} = \frac{i_d}{g_m} + i_d R_S = i_d \left(\frac{1}{g_m} + R_S \right)$$

3. Solve for i_d :

$$i_d = \frac{v_{in}}{\frac{1}{g_m} + R_S}$$

4. The output voltage is $v_{out} = -i_d R_D$:

$$v_{out} = -\frac{v_{in} R_D}{\frac{1}{g_m} + R_S}$$

Voltage Gain with Source Degeneration:

$$A_v = -\frac{R_D}{\frac{1}{g_m} + R_S} = -\frac{g_m R_D}{1 + g_m R_S}$$

Intuition: The gain is the total resistance in the drain divided by the total resistance in the source path (which is $1/g_m + R_S$).

If $R_S \gg 1/g_m$, the gain becomes $A_v \approx -R_D/R_S$. The gain is determined entirely by passive resistors, making it very **linear** and **independent** of the transistor's g_m (which varies with temperature and process).

5.4.2 Output Resistance (with $\lambda \neq 0$)

Source Degeneration Boosts Output Resistance

When we include r_O , the output resistance looking into the drain is significantly **boosted** by the source resistor.

Output Resistance Looking into the Drain:

$$R_{out} \approx r_O(1 + g_m R_S) + R_S \approx (g_m r_O) R_S + r_O$$

The resistor R_S is multiplied by the intrinsic gain of the transistor! This technique (cascode) is widely used to create very high-impedance current sources.

5.4.3 Summary of CS Stages

Configuration	Voltage Gain (A_v)	Output Resistance (R_{out})
Ideal Resistive Load	$-g_m R_D$	R_D
With $\lambda \neq 0$	$-g_m (R_D \parallel r_O)$	$R_D \parallel r_O$
Diode-Connected Load	$-\frac{g_m}{g_m[\text{load}]} = -\sqrt{\frac{(W/L)_1}{(W/L)_{load}}}$	$\frac{1}{g_m[\text{load}]} \parallel r_O[1] \parallel r_O[\text{load}]$
Current-Source Load	$-g_m[1] (r_O[1] \parallel r_O[2])$	$r_O[1] \parallel r_O[2]$
Source Degeneration ($\lambda = 0$)	$-\frac{R_D}{1/g_m + R_S}$	R_D

Table 5.1: Summary of Common-Source Amplifier configurations.

Chapter 6

Practice Problems & Design Challenges

6.1 Fundamentals & I-V Characteristics

Problem 1: Determining the Region of Operation

An NMOS transistor has $\mu_n C_{ox} = 100 \mu\text{A}/\text{V}^2$, $W/L = 10$, and $V_{TH} = 0.5 \text{ V}$. Let $\lambda = 0$. Find the drain current I_D for the following terminal voltages:

- (a) $V_G = 1.5 \text{ V}$, $V_S = 0 \text{ V}$, $V_D = 0.2 \text{ V}$
- (b) $V_G = 1.5 \text{ V}$, $V_S = 0 \text{ V}$, $V_D = 2.0 \text{ V}$
- (c) $V_G = 1.5 \text{ V}$, $V_S = 1.0 \text{ V}$, $V_D = 3.0 \text{ V}$

Solution:

(a) $V_{GS} = 1.5 - 0 = 1.5 \text{ V}$. Since $V_{GS} > V_{TH}$, it is ON. $V_{DS} = 0.2 - 0 = 0.2 \text{ V}$. Overdrive $V_{OV} = V_{GS} - V_{TH} = 1.0 \text{ V}$. Since $V_{DS} < V_{OV}$ ($0.2 < 1.0$), it is in the **Triode Region**.

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[V_{OV} V_{DS} - \frac{V_{DS}^2}{2} \right] = (1000 \mu\text{A}/\text{V}^2) \left[(1.0)(0.2) - \frac{0.04}{2} \right] = 1000 \times 0.18 = 180 \mu\text{A}$$

(b) $V_{GS} = 1.5 \text{ V}$ ($V_{OV} = 1.0 \text{ V}$). $V_{DS} = 2.0 \text{ V}$. Since $V_{DS} > V_{OV}$ ($2.0 > 1.0$), it is in the **Saturation Region**.

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{OV}^2 = \frac{1}{2} (1000 \mu\text{A}/\text{V}^2) (1.0)^2 = 500 \mu\text{A}$$

(c) *Careful: the source is not at 0 V.* $V_{GS} = V_G - V_S = 1.5 - 1.0 = 0.5 \text{ V}$. Since $V_{GS} = V_{TH}$, the channel is just barely formed (edge of conduction). $I_D = 0 \mu\text{A}$ (cutoff / subthreshold).

6.2 Current Mirror Design

Problem 2: Scaling Current Mirrors

Design a current mirror circuit to generate $I_{OUT} = 500 \mu\text{A}$ from a reference current $I_{REF} = 100 \mu\text{A}$. The supply voltage is $V_{DD} = 3.0 \text{ V}$. Assume $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$, $V_{TH} = 0.4 \text{ V}$, and $\lambda = 0$. The reference current is generated by a resistor R tied to V_{DD} .

Requirements: 1. Find $(W/L)_1$ and $(W/L)_2$. 2. The output transistor M_2 must remain in saturation for V_{DS2} as low as 0.2 V . 3. Find the value of R .

Solution:

Step 1: Set the Overdrive Voltage

We are told M_2 must be saturated for $V_{DS2} \geq 0.2 \text{ V}$. Since saturation requires $V_{DS} \geq V_{OV}$, we must design the mirror such that $V_{OV} \leq 0.2 \text{ V}$. Let's choose $V_{OV} = 0.2 \text{ V}$ to minimize device sizes.

Step 2: Size the Output Transistor M_2

For M_2 , $I_{OUT} = 500 \mu\text{A}$ and $V_{OV} = 0.2 \text{ V}$.

$$I_{OUT} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_2 V_{OV}^2$$

$$500 \times 10^{-6} = \frac{1}{2} (200 \times 10^{-6}) \left(\frac{W}{L} \right)_2 (0.2)^2$$

$$500 = 100 \left(\frac{W}{L} \right)_2 (0.04) \Rightarrow 500 = 4 \left(\frac{W}{L} \right)_2$$

$$\left(\frac{W}{L} \right)_2 = 125$$

Step 3: Size the Reference Transistor M_1

The current ratio is $I_{OUT}/I_{REF} = 500/100 = 5$.

$$\frac{(W/L)_2}{(W/L)_1} = 5 \Rightarrow \left(\frac{W}{L} \right)_1 = \frac{125}{5} = 25$$

Step 4: Calculate the Resistor R

The reference device M_1 is diode-connected, so its gate voltage is:

$$V_G = V_{GS} = V_{TH} + V_{OV} = 0.4 + 0.2 = 0.6 \text{ V}$$

The resistor R drops the remaining voltage from V_{DD} :

$$V_{drop} = V_{DD} - V_G = 3.0 - 0.6 = 2.4 \text{ V}$$

$$R = \frac{V_{drop}}{I_{REF}} = \frac{2.4 \text{ V}}{100 \mu\text{A}} = 24 \text{ k}\Omega$$

Summary of Design: $(W/L)_1 = 25$, $(W/L)_2 = 125$, $R = 24 \text{ k}\Omega$.

6.3 Amplifier Analysis

Problem 3: CS Amplifier with Diode-Connected Load

In a CS amplifier with a diode-connected load, M_1 is the amplifying device and M_2 is the diode-connected load. Both are NMOS devices. Given:

- $(W/L)_1 = 50/0.18$
- $(W/L)_2 = 5/0.18$
- $I_D = 1$ mA for both devices
- $\mu_n C_{ox} = 100 \mu\text{A}/\text{V}^2$
- $\lambda = 0.1 \text{ V}^{-1}$ for both devices.

Calculate the exact small-signal voltage gain A_v .

Solution:

Step 1: Calculate g_{m1} and g_{m2}

Using the $g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$ formula:

$$g_{m1} = \sqrt{2(100 \times 10^{-6}) \left(\frac{50}{0.18}\right) (1 \times 10^{-3})} = \sqrt{\frac{10000}{0.18} \times 10^{-9}} = \sqrt{0.0555} \approx 7.45 \text{ mS}$$

$$g_{m2} = \sqrt{2(100 \times 10^{-6}) \left(\frac{5}{0.18}\right) (1 \times 10^{-3})} = \sqrt{\frac{1000}{0.18} \times 10^{-9}} = \sqrt{0.00555} \approx 2.36 \text{ mS}$$

Note: $g_{m1}/g_{m2} = \sqrt{50/5} = \sqrt{10} \approx 3.16$.

Step 2: Calculate r_{O1} and r_{O2}

$$r_{O1} = r_{O2} = \frac{1}{\lambda I_D} = \frac{1}{(0.1)(1 \times 10^{-3})} = 10 \text{ k}\Omega$$

Step 3: Calculate the total load resistance R_{out}

The total resistance at the drain node is $1/g_{m2} \parallel r_{O1} \parallel r_{O2}$.

$$1/g_{m2} = 1/2.36 \text{ mS} = 424 \Omega$$

$$r_{O1} \parallel r_{O2} = 10 \text{ k}\Omega \parallel 10 \text{ k}\Omega = 5 \text{ k}\Omega$$

$$R_{out} = 424 \Omega \parallel 5000 \Omega = \frac{424 \times 5000}{424 + 5000} \approx 391 \Omega$$

Step 4: Calculate the Gain

$$A_v = -g_{m1} R_{out} = -(7.45 \text{ mS})(391 \Omega) = -2.91$$

Comparison to ideal: The ideal formula gives $-\sqrt{(W/L)_1/(W/L)_2} = -\sqrt{10} = -3.16$. The finite output resistance drops the gain slightly to -2.91 .